

A market research between Coke and Mojo

A demand schedule for Coke can be represented by the equation

$$Q_{Dc} = 200 - 5P_c + 2M + 3P_m,$$

where Q_{Dc} is the Quantity Demanded of Coke, P_c is Price of Coke, M is income and P_m is the Price of Mojo.

The supply schedule for Coke can be represented by the equation

$$Q_{Sc} = 100 + 5P_c, \text{ where } Q_{Sc} \text{ is the quantity supplied.}$$

1. Given $M = 80$ and $P_m = 20$, calculate equilibrium Price and Quantity for Coke. [OLD Values]
2. When $M = 100$ and $P_m = 20$, calculate equilibrium Price and Quantity for Coke and then calculate YED of Coke
3. When $M = 80$ and $P_m = 30$, calculate equilibrium Price and Quantity for Coke and then calculate XED of Coke

1) At equilibrium, $Q_{Dc} = Q_{Sc}$

$$\Rightarrow 200 - 5P_c + 2M + 3P_m = 100 + 5P_c$$

$$\Rightarrow 200 - 5P_c + 2(80) + 3(20) = 100 + 5P_c$$

$$\Rightarrow 200 - 5P_c + 160 + 60 = 100 + 5P_c$$

$$\Rightarrow 420 = 10P_c \quad \therefore P_c = \text{Tk. } 42$$

$$\therefore Q_c = 260$$

For elasticity math, the first P & Q values you get will be the old values for the entire math.

2) New equilibrium when $M = 100$, $P_m = \text{Tk. } 20$

$$\Rightarrow 200 - 5P_c + 2M + 3P_m = 100 + 5P_c$$

$$\Rightarrow 200 - 5P_c + 2(100) + 3(20) = 100 + 5P_c$$

$$\Rightarrow 200 - 5P_c + 200 + 60 = 100 + 5P_c$$

$$\therefore P_c = \text{Tk. } 36$$

$$\therefore Q_c = 280$$

$$YED = \frac{\% \Delta \text{ in } Q_c}{\% \Delta \text{ in } I} = \frac{\frac{280 - 260}{\left(\frac{260 + 280}{2}\right)} \times 100}{\frac{100 - 80}{\left(\frac{100 + 80}{2}\right)} \times 100} = \frac{0.0740}{0.2222} = 0.33$$

∴ Coke is a normal good.

3) New equilibrium when $M=80$ $P_M = Tk. 30$

$$\Rightarrow 200 - 5P_C + 2M + 3P_M = 100 + 5P_C$$

$$\Rightarrow 200 - 5P_C + 2(80) + 3(30) = 100 + 5P_C$$

$$\Rightarrow 200 - 5P_C + 160 + 90 = 100 + 5P_C$$

$$\Rightarrow 350 = 10P_C \quad \therefore P_C = Tk. 35$$

$$Q_C = 275$$

$$XED = \frac{\% \Delta \text{ in } Q_C}{\% \Delta \text{ in } P_M} = \frac{\frac{275 - 260}{\frac{275 + 260}{2}} \times 100}{\frac{30 - 20}{\frac{30 + 20}{2}} \times 100} = \frac{0.0561}{0.4} = 0.14$$

∴ Coke and nojo are substitutes.